Skills and Resources Audit

Cover Summary

This audit takes a close look at what skills our team already has and how that compares to what we'll actually need to finish the project. We've listed 20 key skills, from basic programming (JavaScript, HTML, CSS) to more advanced tools and concepts like the Chrome Extension API, YouTube Data API, and fine-tuning large language models.

For each of the skills, we considered how much experience each team member has and designated them as confident, intermediate, basic, or no experience and compared that to how many people we think need to be competent in that skill. We have decided that someone is competent if they have more than basic confidence in that skill. That gives us a quick way to compare where we're strong and where there are gaps.

Whenever we found there was a gap, we decided how we were going to bridge it. It could be through web tutorials, documentation, short training sessions by our team, or through creating smaller practice exercises that feed into the project work. We also made a list of what tools and platforms we were going to use along the way.

The result is an operations manual that not only maps our existing capability but also gives us a formula to fill the gaps so we are able to keep the project on schedule through the different stages.

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1.0 Setup

Skill	Description	Number of people required with at lease intermediate level of skill (leaders)
Basic JavaScript	Extension coding	3
Asynchronous JavaScript	Handle multiple tasks without waiting for each to finish, keeping programs fast and responsive.	2
DOM and content script manipulation	Injecting and modifying the YouTube page content	3
Chrome Extension API and Inter script communication	Allow communication between different parts of the extension	2
Chrome Storage API	Saving user settings	2
Database implementation	Storing large dataset	2
Youtube Data API	Fetching metadata	2
Python	Backend coding and data preprocessing	3
Data preprocessing	Clean and structuring the data/model	3
API integration	Calling LLM endpoints	3
Familiarity with with NLP libraries and frame work	Processing text metadata	3
LLM fine tuning	Classification of user content	3
Allostatic regulation model (Client formula)	For adaptive filter implementation	4
HTML	Dashboard layout	3
CSS + bootstrap/tailwind	Dashboard/UI styling	3
Figma	UI/UX design	2
Unit test - javascript	Frontend test	2
Unit test - python	Backend test	2
All GitHub and git operations/actions	Version control and collaboration	4
Documentation - Readme file / setup file	Proper documentation	3

2.0 Detailed Individual Assessment

Leaders we have	Skill	Kean	Keandria	Luvish	Mashudul	Sia	Tanish
5	Basic JavaScript	Intermediate	Intermediate	Confident	Basic	Confident	Confident
4	Asynchronous JavaScript	Intermediate	Intermediate	Basic	Basic	Intermediate	Intermediate
3	DOM and content script manipulation	Intermediate	No experience	Intermediate	No experience	Confident	No experience
0	Chrome Extension API and Inter script communication	No experience	No experience	No experience	No experience	Basic	Basic
0	Chrome Storage API	No experience	No experience	No experience	No experience	Basic	No experience
4	Database implementation	Intermediate	Confident	Confident	No experience	Confident	No experience
1	Youtube Data API	Intermediate	No experience	No experience	No experience	Basic	No experience
5	Python	Confident	Confident	Confident	Basic	Intermediate	Confident
3	Data preprocessing	Confident	Basic	Confident	No experience	Confident	No experience
2	API integration	Intermediate	No experience	Intermediate	No experience	Basic	Basic
1	Familiarity with with NLP libraries and frame work	Intermediate	No experience	No experience	No experience	Basic	No experience
0	LLM fine tuning	Basic	No experience	No experience	No experience	Basic	No experience
0	Allostatic regulation model (Client formula)	Basic	No experience	Basic	No experience	Basic	Basic
5	HTML	Basic	Confident	Confident	Intermediate	Confident	Confident
4	CSS + bootstrap/tailwind	Basic	Confident	Confident	Basic	Intermediate	Intermediate
2	Figma	No experience	Confident	Confident	Basic	Basic	Basic
0	Unit test - javascript	Basic	No experience	Basic	No experience	Basic	No experience
3	Unit test - python	Intermediate	Intermediate	Basic	No experience	Intermediate	No experience
5	All github and git operations/actions	Intermediate	Confident	Confident	Confident	Confident	No experience
4	Documentation - Readme file / setup file	Intermediate	Confident	Confident	Basic	Confident	No experience

3.0 Skill Summary and Gap Analysis

Number of leaders needed	Skill	Confident	Intermediate	Basic	No Experience	Gap
3	Basic JavaScript	3	2	1	0	No
2	Asynchronous JavaScript	0	4	2	0	No
3	DOM and content script manipulation	1	2	0	3	No
2	Chrome Extension API and Inter script communication	0	0	2	4	Yes
2	Chrome Storage API	0	0	1	5	Yes
2	Database implementation	3	1	0	2	No
2	Youtube Data API	0	1	1	4	Yes
3	Python	4	1	1	0	No
3	Data preprocessing	3	0	1	2	No
2	API integration	0	2	2	2	No
3	Familiarity with NLP libraries and frame work	0	1	1	4	Yes
3	LLM fine tuning	0	0	2	4	Yes
4	Allostatic regulation model (Client formula)	0	0	4	2	Yes
3	HTML	4	1	1	0	No
3	CSS + bootstrap/tailwind	2	2	2	0	No
2	Figma	2	0	3	1	No
2	Unit test - javascript	0	0	3	3	Yes
2	Unit test - python	0	3	1	2	No
4	All github and git operations/actions	4	1	0	1	No
3	Documentation - Readme file / setup file	3	1	1	1	No

4.0 Gap Mitigation Plan

All members will develop the skills outlined below as the project progresses. However, owners will bear the primary responsibility for learning and applying them.

Skill Gap	Impact	Mitigation	Owner	Due	Risk and Contingency
Chrome Extension API and Inter script communication	Cannot build extension	Follow tutorials on DEV Community and follow/read through the Chrome Extension Developer guide.	Luvish Keandria	Week 5	Leverage existing open-source extensions for reference to maintain progress while gaining proficiency.
Chrome Storage API	Cannot save user settings	Follow tutorials on DEV Community and follow/read through the Chrome Extension Developer guide.	Luvish Keandria	Week 5	Use alternative data storage
Youtube Data API	Cannot fetch video metadata	Checkout the Google API quick start guides and YouTube Data API documentations.	Mashudul Tanish	Week 5	If API quota limits usage will have to be monitored
Familiarity with NLP libraries and frame work	Cannot process metadata	Follow tutorials for NLP on GeeksforGeeks.	Kean Sia Luvish	Week 6	Use pre-trained models to reduce training need
LLM fine tuning	Cannot customise model to fit client's need	Research model fine turning techniques.	Kean Sia Keandria	Week 6	Use pre trained model until fine tuning possible.
Allostatic regulation model	No dynamic filtering in accordance to client's research	Read and understand mathematical formula in client's academic paper.	Kean Sia Luvish Keandria	Week 5	Use a liner model or a rule based model
Unit test - javascript	No automated frontend tests	Learn one of JavaScript's testing framework and write tests for understanding.	Mashudul Tanish	Week 6	Temporarily carry manual testing.

5.0 Resource Audit

Resource	Purpose	Status	Action/Notes
GitHub Page	Host the documents of the extension	Need	For setup instructions and documentation.
GitHub repository	Version control	Have	Private now, turn public after development.
Figma	Dashboard/UI/UX design	Have	Group project has been set up for the development of dashboard.
LLM API access - OpenAI (GPT-4.1 Nano)	Classification of user content	Need	Awaiting funding approval. If not approved, will use GPT-4.1 Mini
Testing Framework	Carry out unit testing	Need	Needs to be installed.
YouTube API Access	Access metadata	Need	Access through Google Cloud Console.
Chrome Browser - latest version	Development	Have	Everyone has the latest chrome version.
Laptop	Development	Have	Everyone has a laptop.
IDE	Development	Have	Everyone has a different IDE.